# Neural Networks

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# Agenda

- Concept Recap
- Hands-On Demonstration

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# Let's begin the discussion by answering a few questions on neural networks

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# Neural Networks Discussion Question

**A neuron receives two inputs, $x_1$ and $x_2$. Which sequence correctly describes how the neuron produces its output?**

A. Multiply inputs by their weights, add the bias, then apply an activation function to the result

B. Apply an activation function to each input, then multiply by weights and add the bias

C. Add the bias to each input first, then multiply by weights, skipping the activation function

D. Multiply inputs by their weights and sum them - this sum directly becomes the neuron's output

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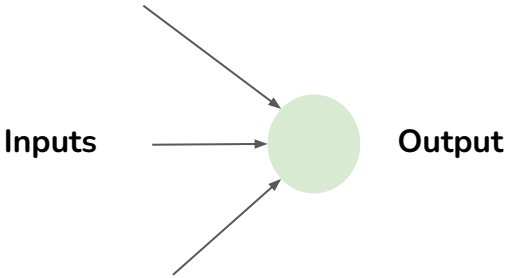
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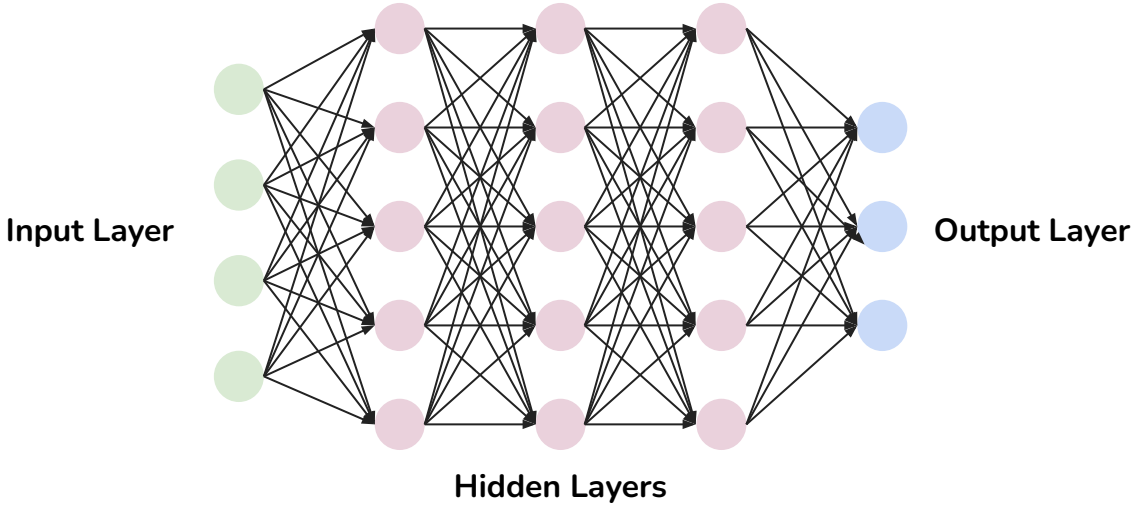
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# Neural Network Architecture

A neural network architecture comprising interconnected input, hidden, and output layers, facilitating the learning of complex relationships between input and output data




Neuron



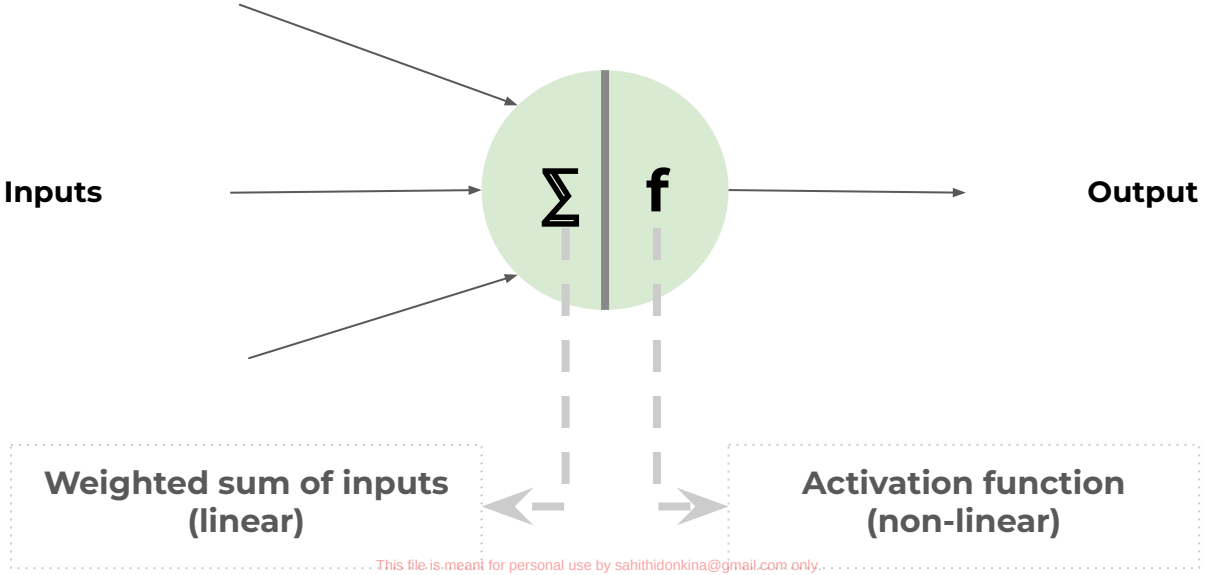

Neural Network

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# Activation Functions

Determines the output of a neuron based on the input

**Introduces non-linearity** and enables the neural network to learn complex patterns in the data




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# Neural Networks Discussion Question

**Consider a neural network built to predict whether a student will pass or fail in an exam. Which of the following activation functions would be most appropriate to use in the output layer?**

A. Sigmoid

B. Softmax

C. Linear

D. Mean Squared Error

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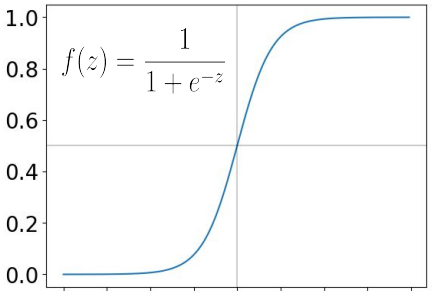
C. Linear

D. Mean Squared Error

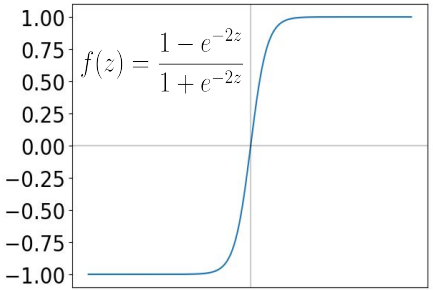
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# Activation Functions for Hidden Layers

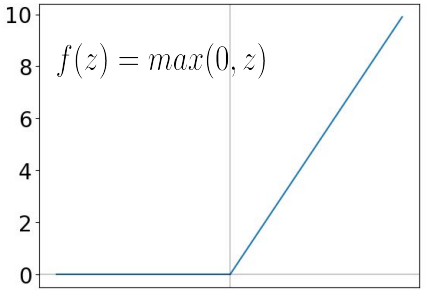
**Sigmoid**




**Tanh**

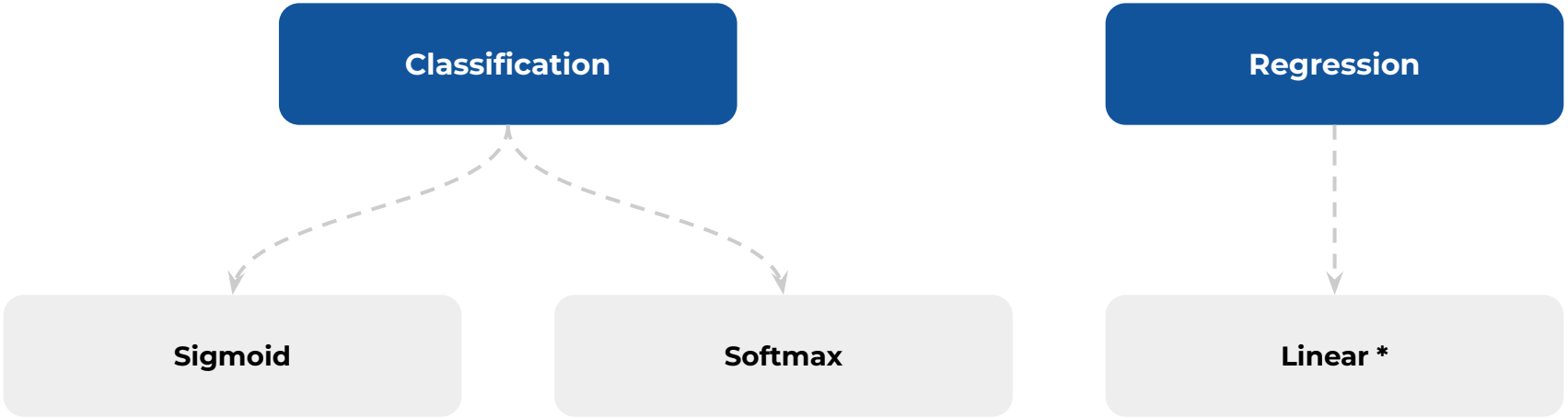



**ReLU**




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# Activation Functions for Output Layer




* This means no activation is applied in the output layer

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# Neural Networks Discussion Question

**When training a neural network, how does Adam typically behave differently from SGD on the loss curve?**

A. Adam takes uniform, fixed-size steps at every point on the curve, just like SGD

B. Adam adapts its step size based on past gradients, often converging faster and with less oscillation than SGD

C. Adam and SGD always reach the same minimum in the same number of steps

D. Adam only works for convex loss surfaces, while SGD works for all loss surfaces

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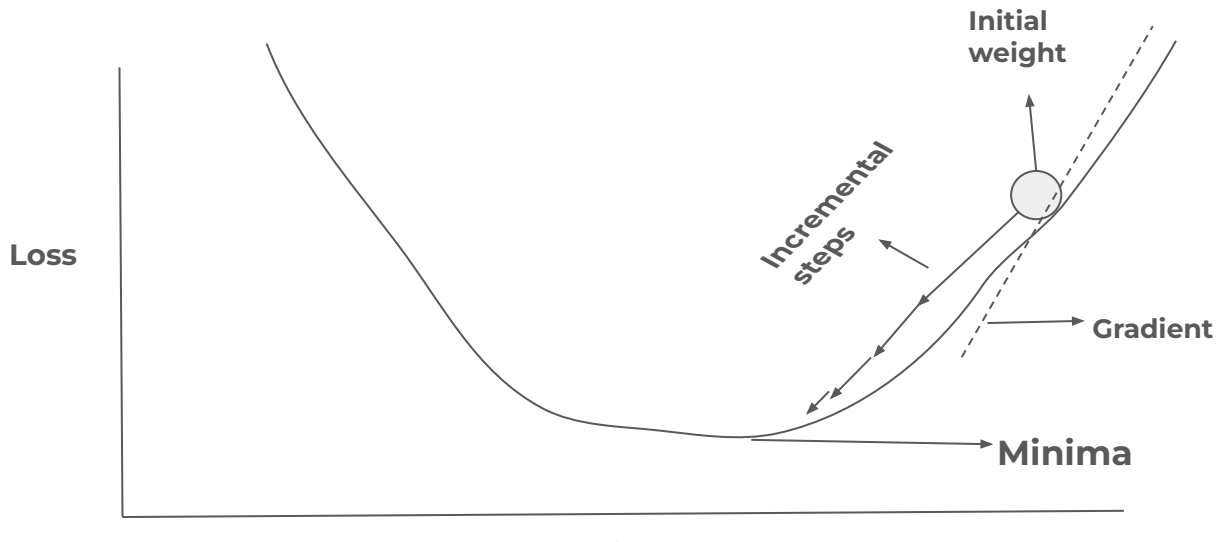
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# Gradient Descent

An **optimization algorithm** used to minimize the loss function

**Adjusts model parameters iteratively** in the direction of steepest descent of the gradient




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# SGD (Stochastic Gradient Descent) vs Adam

| SGD | Adam |
| --- | --- |
| Stochastic Gradient Descent | Adaptive Moment Estimation |
| No momentum component | Adds a momentum component |
| Fixed Learning Rate | Adaptive Learning Rate |

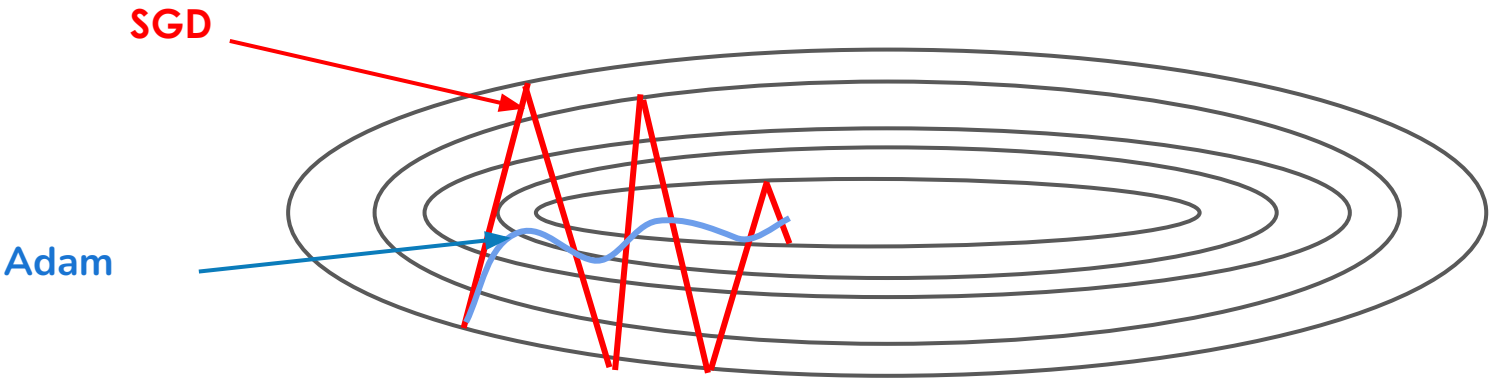
The learning rate is different for each model parameter and depends on the value of the gradient

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# SGD (Stochastic Gradient Descent) vs Adam

SGD's step size is the learning rate times the current gradient - with no memory of past steps and no per-parameter adjustment. On uneven loss surfaces, this leads to oscillation and slower convergence.

Adam combines a momentum-like term (smooths direction using past gradients) with an adaptive learning rate (scales each parameter step individually) - resulting in faster, smoother convergence than SGD.




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# Neural Networks Discussion Question

**During backpropagation, which of the following parameters of the model gets updated?**

A. Input layer parameters

B. Output layer parameters

C. Hidden layer parameters

D. All trainable model parameters

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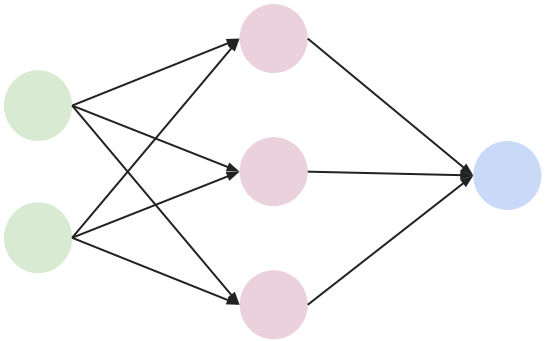
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# Training Neural Networks - Backpropagation

An **algorithm** used to efficiently compute gradients of the loss function with respect to model parameters



Input layer Hidden layer Output layer

**Forward Propagation**

Gradient of error is calculated with respect to each weight

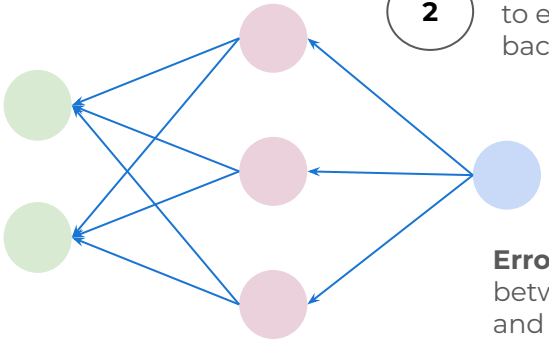
3

2

Error is sent back to each neuron in backward direction

1

**Error -** difference between predicted and actual output



Input layer Hidden layer Output layer

**Backpropagation**

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# Neural Networks Discussion Question

**Which of the following regularization techniques involve trainable parameters?**

A. Dropout

B. Batch Normalization

C. Both Dropout and Batch Normalization

D. Neither Dropout nor Batch Normalization

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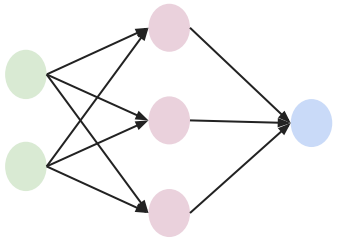
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# Dropout

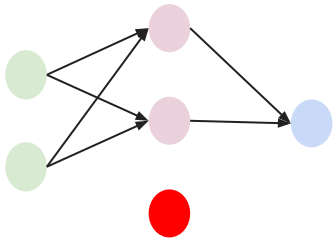
**Dropout prevents overfitting** by randomly deactivating neurons during training

**Independence:** Fosters independent learning among neurons

**Ensemble:** Creates a diverse ensemble of networks as neurons are deactivated randomly during training and the final result is an average prediction



Before Dropout



After Dropout

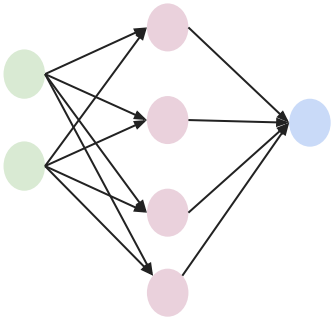
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# Dropout

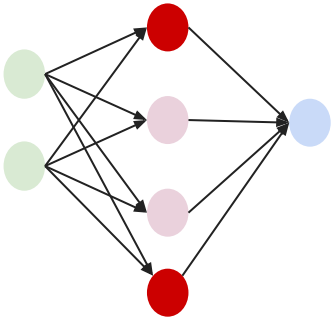
**Step 1 -** Choose a dropout rate p (p = 0.5 here)

**Step 2 -** Randomly select 100*p% of the neurons (50% here)

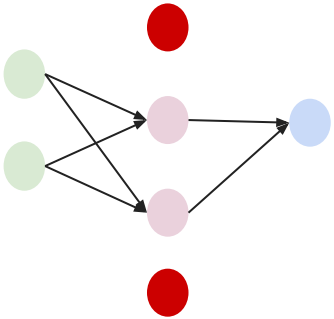
**Step 3 -** Deactivate the selected neurons by setting them to zero.



1



2



3

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# Batch Normalization

BatchNorm stabilizes training primarily by smoothing the optimization landscape, making gradients more predictable and allowing larger learning rates.

**Regularization:** Adds slight noise via mini-batch statistics, acting as a mild regularizer.

**Formula:** $y_i = \gamma \hat{x}_i + \beta$

where $\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$

(subtract mean, then divide by standard deviation to normalise)

**Mean:** $\mu_B = \frac{1}{m}\sum_{i=1}^{m} x_i$

**Standard Deviation:** $\sigma_B^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_B)^2$

**Epsilon $\epsilon$** : Prevents division by zero when batch variance is very small, keeping the computation numerically stable.

**Two trainable parameters: γ** (scale) & **β** (shift)

**Trainable Parameters (γ, β):** Learned during training via backpropagation - lets the network scale/shift the normalized values, so normalization never limits what the layer can learn.

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# Power Ahead!



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